3D Website Building Guide - AI Prompts & Setup

🎨 AI Prompts for Frame Generation

1. Starting Frame

Use this for: Initial calm state of your product

A photorealistic studio product render of two disposable takeaway coffee cups floating against a warm chocolate-brown gradient background. The composition is wide with large negative space on the left. On the right, one darker brown coffee cup stands upright with a white plastic lid and a ribbed cardboard sleeve. Slightly in front and angled is a lighter cream-colored cup with subtle speckled texture, a black lid, and a smooth kraft paper sleeve. Soft cinematic studio lighting, gentle highlights, realistic paper textures, shallow depth of field, cozy café atmosphere, premium minimalist aesthetic, ultra-detailed, high resolution, no text, no logos, no people.

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2. Ending Frame

Use this for: Final dramatic state with action

A hyper-realistic studio product render of two disposable takeaway coffee cups floating against a warm chocolate-brown gradient background with large negative space on the left. On the right, a darker brown cup stands upright with a white plastic lid and ribbed cardboard sleeve. In front, a cream-colored cup with subtle speckled texture and a smooth kraft paper sleeve is tilted diagonally. Its black lid is lifted mid-air as a dramatic splash of hot coffee erupts from the cup, frozen in motion with glossy caramel-brown liquid, dynamic arcs, and suspended droplets. Cinematic studio lighting, high-speed splash photography style, ultra-detailed paper textures, shallow depth of field, premium café advertising aesthetic, no text, no logos, no people, high resolution.

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3. Animation Sequence

Use this for: Creating 60-120 transition frames between start and end

A high-end cinematic product animation transitioning between two provided keyframes. Wide horizontal frame with a locked camera and warm chocolate-brown gradient background. On the right side, two disposable takeaway coffee cups float in mid-air. The animation begins perfectly matching the first frame: both cups calm and still, one dark brown cup with white lid and ribbed sleeve behind, one cream-colored speckled cup with kraft sleeve in front. As the animation progresses, the cream cup slowly tilts forward, its black lid lifting upward in a

smooth arc. A rich caramel-brown coffee liquid erupts from the cup with realistic viscosity, forming elegant splash arcs and suspended droplets. Motion is smooth, slow, and premium, inspired by high-speed liquid photography but gracefully decelerated. The animation ends perfectly matching the final frame: lid lifted, splash frozen mid-air, droplets suspended, cinematic studio lighting, shallow depth of field, ultra-realistic materials, premium café advertising aesthetic, no text, no logos, no people.

🛠️ Tech Stack

- **Framework:** Next.js 14 (App Router)
- **Language:** TypeScript
- **Styling:** Tailwind CSS
- **Animation:** Framer Motion + HTML5 Canvas
- **Fonts:** Inter (body) + Playfair Display (headings)

✅ ACT AS

A world-class Creative Developer (Awwwards-level) specializing in:

- Next.js 14+ (App Router)
- TypeScript
- Tailwind CSS
- Framer Motion
- Scroll-triggered HTML5 Canvas animations
- Premium coffee shop web experiences

🎯 THE TASK

Build a complete, production-ready **premium coffee brand website** with:

HERO SECTION (100vh):

Scroll-triggered frame-by-frame canvas animation with:

- **Bi-directional scroll control** (plays forward on scroll down, reverses on scroll up)
- **Anti-gravity physics effect** (subtle upward float with spring motion)
- **Sticky canvas** that stays fixed while user scrolls through animation
- User-provided frames from `/public/frames/` directory

PRODUCT SECTIONS (Below Hero):

Coffee product showcase following the uploaded reference image design:

- Coffee product cards with ratings, prices, add-to-cart buttons
- Feature highlight sections with coffee imagery
- Warm brown coffee shop aesthetic
- Smooth scroll reveals with Framer Motion

Reference Website: https://lighthearted-conkies-6bf67a.netlify.app/ (for scroll animation inspiration)

🧰 TECH STACK

- **Framework:** Next.js 14 (App Router)
- **Language:** TypeScript (100% type-safe)
- **Styling:** Tailwind CSS
- **Animation:** Framer Motion + HTML5 Canvas
- **Fonts:** Google Fonts ('Inter' for body, 'Playfair Display' for headings)
- **Deployment:** Static Export compatible (Netlify/Vercel)

🎨 VISUAL DIRECTION & COLOR PALETTE

Hero Section (Canvas Animation):

- Background: `#1A0F0A` (deep espresso black-brown)
- Canvas must blend seamlessly with background
- Text overlays: `text-amber-50/95` with subtle glow
- Anti-gravity particles: floating coffee beans with `#D4A574` tint

Product Sections (From Reference Image):

```
--coffee-bg-primary: #2D1810;     /* Rich coffee brown */
--coffee-bg-secondary: #3D2820;   /* Card backgrounds */
--coffee-border: #5A4034;         /* Subtle borders */
--coffee-text-primary: #F5E6D3;   /* Cream/latte text */
--coffee-text-secondary: #C9B8A0; /* Muted brown */
--coffee-accent: #4F9C8F;         /* Teal CTA buttons */
--coffee-gold: #FFD700;           /* Star ratings */
--coffee-gradient: linear-gradient(135deg, #2D1810 0%, #1A0F0A 100%);
```

Typography:

- **Headings:** `'Playfair Display'` (elegant serif) - weights 400, 600, 700
- **Body:** `'Inter'` (clean sans-serif) - weights 400, 500, 600
- **Hero Title:** `text-8xl md:text-9xl` with tight tracking

- **Product Cards:** `text-xl` titles, `text-sm` descriptions

📂 PROJECT STRUCTURE & FILES TO GENERATE

You must generate **COMPLETE, PRODUCTION-READY** files:

A. Configuration Files

`next.config.mjs`

```
/** @type {import('next').NextConfig} */
const nextConfig = {
  output: 'export',
  images: {
    unoptimized: true
  }
};

export default nextConfig;
```

`tailwind.config.ts`

- Extend colors with coffee palette
- Add custom animations: `float`, `glow`, `slide-up`
- Configure backdrop-blur utilities

`app/globals.css`

```
@import
url('https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600&family=Playfair+Display:wght@400;600;700&display=swap');

@tailwind base;
@tailwind components;
@tailwind utilities;

@layer base {
  body {
    @apply bg-[#1A0F0A] text-[#F5E6D3] font-['Inter'] antialiased;
  }

  ::selection {
    @apply bg-[#4F9C8F] text-white;
  }

  /* Custom Scrollbar */
```

```
  ::-webkit-scrollbar {
    width: 10px;
  }

  ::-webkit-scrollbar-track {
    background: #1A0F0A;
  }

  ::-webkit-scrollbar-thumb {
    background: #5A4034;
    border-radius: 5px;
  }

  ::-webkit-scrollbar-thumb:hover {
    background: #7A5044;
  }
}
```

B. Layout & Metadata

app/layout.tsx

```
import type { Metadata } from 'next';
import { Inter, Playfair_Display } from 'next/font/google';
import './globals.css';

const inter = Inter({
  subsets: ['latin'],
  variable: '--font-inter'
});

const playfair = Playfair_Display({
  subsets: ['latin'],
  variable: '--font-playfair'
});

export const metadata: Metadata = {
  title: 'Artisan Coffee | Experience Excellence',
  description: 'Premium coffee experiences crafted to perfection',
};

export default function RootLayout({
  children,
}: {
  children: React.ReactNode;
}) {
```

```
  return (
    <html lang="en" className={`${inter.variable} ${playfair.variable}`}>
      <body>{children}</body>
    </html>
  );
}
```

C. Data Layer

`data/products.ts`

```
export interface CoffeeProduct {
  id: string;
  name: string;
  description: string;
  price: string;
  rating: number;
  image: string;
  features: string[];
}

export const coffeeProducts: CoffeeProduct[] = [
  {
    id: 'cappuccino',
    name: 'Cappuccino',
    description: 'Cappuccino is a latte made with more foam than steamed milk, often topped
with cocoa powder.',
    price: '$3.50',
    rating: 4.9,
    image: '/coffee/cappuccino.jpg',
    features: ['Espresso', 'Steamed Milk', 'Foam']
  },
  {
    id: 'latte',
    name: 'Latte',
    description: 'Latte is a coffee drink made with espresso and steamed milk. Rich, creamy,
balanced.',
    price: '$4.00',
    rating: 5.0,
    image: '/coffee/latte.jpg',
    features: ['Espresso', 'Steamed Milk', 'Light Foam']
  },
  {
    id: 'mocha',
    name: 'Mocha',
```

```
    description: 'Mocha is a coffee beverage where dark espresso meets rich chocolate and
creamy milk.',
    price: '$4.50',
    rating: 4.7,
    image: '/coffee/mocha.jpg',
    features: ['Espresso', 'Chocolate', 'Steamed Milk']
  }
];

export interface FeatureHighlight {
  title: string;
  description: string;
  position: 'left' | 'right';
}

export const features: FeatureHighlight[] = [
  {
    title: 'High-Quality Beans',
    description: 'High-quality beans are a single told story about craft, dedication, and the
culinary journey where every sip is unique.',
    position: 'left'
  },
  {
    title: 'Individual Approach',
    description: 'Most visitors expect coffee culture today is meticulously designed and
economically managed. Individual approach.',
    position: 'right'
  },
  {
    title: 'Atmosphere of Inspiration',
    description: 'Lorem ipsum is dolor sit amet, consectetur adipiscing elit. Suspendisse
dapibus tempor incididunt ut.',
    position: 'left'
  },
  {
    title: 'Professional Baristas',
    description: 'Professional Baristas are server professional and deliver rich experiences
with every custom coffee crafted perfectly.',
    position: 'right'
  }
];
```

D. Core Components

`components/HeroCanvasAnimation.tsx`

THIS IS THE MAIN ENGINE - CRITICAL IMPLEMENTATION:

```
'use client';

import { useRef, useEffect, useState } from 'react';
import { motion, useScroll, useTransform, useSpring, useVelocity } from 'framer-motion';

const TOTAL_FRAMES = 120; // Adjust based on your frame count
const FRAME_PATH = '/frames'; // Folder containing frame_0.webp to frame_119.webp

export default function HeroCanvasAnimation() {
  const containerRef = useRef<HTMLDivElement>(null);
  const canvasRef = useRef<HTMLCanvasElement>(null);

  const [images, setImages] = useState<HTMLImageElement[]>([]);
  const [imagesLoaded, setImagesLoaded] = useState(false);
  const [loadProgress, setLoadProgress] = useState(0);

  // Scroll progress tracking
  const { scrollYProgress } = useScroll({
    target: containerRef,
    offset: ['start start', 'end start']
  });

  // Smooth spring animation for buttery scroll
  const smoothProgress = useSpring(scrollYProgress, {
    stiffness: 100,
    damping: 30,
    restDelta: 0.001
  });

  // Anti-gravity effect based on scroll velocity
  const scrollVelocity = useVelocity(scrollYProgress);
  const yOffset = useTransform(
    scrollVelocity,
    [-1, 0, 1],
    [15, 0, -15] // Floats up when scrolling down
  );

  // Map scroll to frame index (bi-directional)
  const frameIndex = useTransform(
    smoothProgress,
    [0, 1],
    [0, TOTAL_FRAMES - 1]
  );

  // Preload all frames
  useEffect(() => {
```

```
  const loadImages = async () => {
    const imagePromises = Array.from({ length: TOTAL_FRAMES }, (_, i) => {
      return new Promise<HTMLImageElement>((resolve, reject) => {
        const img = new Image();
        img.src = `${FRAME_PATH}/frame_${i}.webp`;
        img.onload = () => {
          setLoadProgress((prev) => prev + (100 / TOTAL_FRAMES));
          resolve(img);
        };
        img.onerror = reject;
      });
    });

    const loadedImages = await Promise.all(imagePromises);
    setImages(loadedImages);
    setImagesLoaded(true);
  };

  loadImages();
}, []);

// Canvas rendering
useEffect(() => {
  if (!imagesLoaded || !canvasRef.current) return;

  const canvas = canvasRef.current;
  const ctx = canvas.getContext('2d');
  if (!ctx) return;

  const renderFrame = () => {
    const currentFrame = Math.round(frameIndex.get());
    const img = images[Math.max(0, Math.min(currentFrame, TOTAL_FRAMES - 1))];

    if (img) {
      // Responsive canvas sizing
      canvas.width = window.innerWidth;
      canvas.height = window.innerHeight;

      // Calculate scaling (contain fit)
      const scale = Math.min(
        canvas.width / img.width,
        canvas.height / img.height
      );

      const x = (canvas.width - img.width * scale) / 2;
      const y = (canvas.height - img.height * scale) / 2;

      ctx.clearRect(0, 0, canvas.width, canvas.height);
```

```
      ctx.drawImage(img, x, y, img.width * scale, img.height * scale);
    }
  };

  const unsubscribe = frameIndex.on('change', renderFrame);
  renderFrame(); // Initial render

  // Handle window resize
  const handleResize = () => renderFrame();
  window.addEventListener('resize', handleResize);

  return () => {
    unsubscribe();
    window.removeEventListener('resize', handleResize);
  };
}, [imagesLoaded, images, frameIndex]);

// Text overlay animations
const section1Opacity = useTransform(smoothProgress, [0, 0.1, 0.2, 0.25], [0, 1, 1, 0]);
const section2Opacity = useTransform(smoothProgress, [0.3, 0.35, 0.5, 0.55], [0, 1, 1, 0]);
const section3Opacity = useTransform(smoothProgress, [0.6, 0.65, 0.8, 0.85], [0, 1, 1, 0]);
const section4Opacity = useTransform(smoothProgress, [0.9, 0.92, 0.98, 1], [0, 1, 1, 0]);

const scrollIndicatorOpacity = useTransform(smoothProgress, [0, 0.1], [1, 0]);

if (!imagesLoaded) {
  return (
    <div className="fixed inset-0 bg-[#1A0F0A] flex flex-col items-center justify-center
z-50">
      <div className="w-64 h-2 bg-amber-900/30 rounded-full overflow-hidden mb-4">
        <motion.div
          className="h-full bg-gradient-to-r from-[#D4A574] to-[#4F9C8F]"
          initial={{ width: '0%' }}
          animate={{ width: `${loadProgress}%` }}
          transition={{ duration: 0.3 }}
        />
      </div>
      <p className="text-amber-100/70 text-lg font-['Inter']">
        Loading Experience... {Math.round(loadProgress)}%
      </p>
    </div>
  );
}

return (
  <div ref={containerRef} className="relative h-[500vh]">
    <div className="sticky top-0 h-screen w-full overflow-hidden">
      <motion.div style={{ y: yOffset }} className="w-full h-full">
```

```
        <canvas
          ref={canvasRef}
          className="w-full h-full"
        />
      </motion.div>

      {/* Text Overlays */}
      <div className="absolute inset-0 pointer-events-none flex items-center justify-center">
        <motion.div
          style={{ opacity: section1Opacity }}
          className="text-center px-4"
        >
          <h1 className="text-7xl md:text-9xl font-['Playfair_Display'] font-bold text-amber-50
mb-4 tracking-tight">
            Experience Coffee
          </h1>
          <p className="text-xl md:text-2xl text-amber-100/80 font-['Inter']">
            Where every sip defies gravity
          </p>
        </motion.div>

        <motion.div
          style={{ opacity: section2Opacity }}
          className="text-left px-8 md:px-16 max-w-2xl"
        >
          <h2 className="text-5xl md:text-7xl font-['Playfair_Display'] font-semibold
text-amber-50 mb-3">
            Crafted to Perfection
          </h2>
          <p className="text-lg md:text-xl text-amber-100/70 font-['Inter']">
            From bean to cup, excellence floats in every drop
          </p>
        </motion.div>

        <motion.div
          style={{ opacity: section3Opacity }}
          className="text-right px-8 md:px-16 max-w-2xl ml-auto"
        >
          <h2 className="text-5xl md:text-7xl font-['Playfair_Display'] font-semibold
text-amber-50 mb-3">
            Anti-Gravity Flavor
          </h2>
          <p className="text-lg md:text-xl text-amber-100/70 font-['Inter']">
            Defying expectations, elevating taste beyond limits
          </p>
        </motion.div>

        <motion.div
```

```
          style={{ opacity: section4Opacity }}
          className="text-center px-4"
        >
          <h2 className="text-6xl md:text-8xl font-['Playfair_Display'] font-bold text-amber-50
mb-6">
            Discover Your Blend
          </h2>
          <motion.button
            whileHover={{ scale: 1.05 }}
            whileTap={{ scale: 0.95 }}
            className="px-8 py-4 bg-gradient-to-r from-[#4F9C8F] to-[#3D8B7F] text-white
rounded-full text-lg font-semibold shadow-2xl pointer-events-auto"
          >
            Explore Collection ↓
          </motion.button>
        </motion.div>
      </div>

      {/* Scroll Indicator */}
      <motion.div
        style={{ opacity: scrollIndicatorOpacity }}
        className="absolute bottom-12 left-1/2 -translate-x-1/2 flex flex-col items-center
gap-2"
      >
        <p className="text-amber-100/60 text-sm font-['Inter'] tracking-wider uppercase">
          Scroll to Explore
        </p>
        <motion.div
          animate={{ y: [0, 8, 0] }}
          transition={{ repeat: Infinity, duration: 1.5 }}
          className="w-6 h-10 border-2 border-amber-100/40 rounded-full flex items-start
justify-center p-2"
        >
          <div className="w-1 h-3 bg-amber-100/60 rounded-full" />
        </motion.div>
      </motion.div>
    </div>
  </div>
);
}
```

---

components/ProductCard.tsx

```
'use client';

import { motion } from 'framer-motion';
```

```
import { CoffeeProduct } from '@/data/products';

interface ProductCardProps {
  product: CoffeeProduct;
  index: number;
}

export default function ProductCard({ product, index }: ProductCardProps) {
  return (
    <motion.div
      initial={{ opacity: 0, y: 50 }}
      whileInView={{ opacity: 1, y: 0 }}
      viewport={{ once: true, margin: '-100px' }}
      transition={{ duration: 0.6, delay: index * 0.1 }}
      whileHover={{ scale: 1.05, y: -5 }}
      className="bg-[#3D2820]/80 backdrop-blur-sm rounded-2xl p-6 border
border-[#5A4034] hover:border-[#4F9C8F] transition-all duration-300 shadow-xl
hover:shadow-2xl hover:shadow-[#4F9C8F]/20"
    >
      {/* Star Rating */}
      <div className="flex items-center gap-2 mb-4">
        <span className="text-[#FFD700] text-lg">★</span>
        <span className="text-[#F5E6D3] font-semibold text-sm">{product.rating}</span>
      </div>

      {/* Coffee Image */}
      <div className="w-full h-56 bg-[#2D1810] rounded-xl mb-5 overflow-hidden">
        <img
          src={product.image}
          alt={product.name}
          className="w-full h-full object-cover hover:scale-110 transition-transform
duration-500"
        />
      </div>

      {/* Title & Description */}
      <h3 className="text-2xl font-['Playfair_Display'] font-bold text-[#F5E6D3] mb-3">
        {product.name}
      </h3>
      <p className="text-sm text-[#C9B8A0] mb-5 line-clamp-2 font-['Inter']">
        {product.description}
      </p>

      {/* Price & Add Button */}
      <div className="flex items-center justify-between">
        <span className="text-3xl font-bold text-[#F5E6D3] font-['Inter']">
          {product.price}
        </span>
```

```
        <motion.button
          whileHover={{ scale: 1.1, rotate: 90 }}
          whileTap={{ scale: 0.9 }}
          className="w-12 h-12 rounded-full bg-gradient-to-br from-[#4F9C8F] to-[#3D8B7F]
flex items-center justify-center hover:shadow-lg hover:shadow-[#4F9C8F]/40
transition-shadow"
        >
          <span className="text-white text-2xl font-bold">+</span>
        </motion.button>
      </div>
    </motion.div>
  );
}
```

---

components/FeatureSection.tsx

```
'use client';

import { motion } from 'framer-motion';
import { features } from '@/data/products';

export default function FeatureSection() {
  return (
    <section className="py-24 px-4 md:px-8 relative overflow-hidden">
      {/* Background Gradient */}
      <div className="absolute inset-0 bg-gradient-to-b from-[#1A0F0A] via-[#2D1810]
to-[#1A0F0A] opacity-50" />

      <div className="max-w-7xl mx-auto relative z-10">
        <div className="grid grid-cols-1 lg:grid-cols-3 gap-12 items-center">
          {/* Left Features */}
          <div className="space-y-8">
            {features.filter(f => f.position === 'left').map((feature, i) => (
              <motion.div
                key={i}
                initial={{ opacity: 0, x: -50 }}
                whileInView={{ opacity: 1, x: 0 }}
                viewport={{ once: true }}
                transition={{ duration: 0.6, delay: i * 0.2 }}
                className="bg-[#3D2820]/60 backdrop-blur-sm p-6 rounded-xl border
border-[#5A4034]/50"
              >
                <h3 className="text-2xl font-['Playfair_Display'] font-semibold text-[#F5E6D3]
mb-3">
                  {feature.title}
                </h3>
```

```
            <p className="text-sm text-[#C9B8A0] font-['Inter'] leading-relaxed">
              {feature.description}
            </p>
          </motion.div>
        ))}
      </div>

      {/* Center: Coffee Cup Image */}
      <motion.div
        initial={{ opacity: 0, scale: 0.8 }}
        whileInView={{ opacity: 1, scale: 1 }}
        viewport={{ once: true }}
        transition={{ duration: 0.8 }}
        className="flex items-center justify-center"
      >
        <div className="relative">
          <motion.div
            animate={{ rotate: 360 }}
            transition={{ repeat: Infinity, duration: 20, ease: 'linear' }}
            className="absolute inset-0 bg-gradient-to-r from-[#4F9C8F]/20 to-[#D4A574]/20
rounded-full blur-3xl"
          />
          <img
            src="/coffee/cup-centered.png"
            alt="Premium Coffee Cup"
            className="relative z-10 w-80 h-80 object-contain drop-shadow-2xl"
          />
        </div>
      </motion.div>

      {/* Right Features */}
      <div className="space-y-8">
        {features.filter(f => f.position === 'right').map((feature, i) => (
          <motion.div
            key={i}
            initial={{ opacity: 0, x: 50 }}
            whileInView={{ opacity: 1, x: 0 }}
            viewport={{ once: true }}
            transition={{ duration: 0.6, delay: i * 0.2 }}
            className="bg-[#3D2820]/60 backdrop-blur-sm p-6 rounded-xl border
border-[#5A4034]/50"
          >
            <h3 className="text-2xl font-['Playfair_Display'] font-semibold text-[#F5E6D3]
mb-3">
              {feature.title}
            </h3>
            <p className="text-sm text-[#C9B8A0] font-['Inter'] leading-relaxed">
              {feature.description}
```

```
            </p>
          </motion.div>
        ))}
      </div>
    </div>
  </div>
 </section>
 );
}
```

---

components/ProductShowcase.tsx

```
'use client';

import { motion } from 'framer-motion';
import ProductCard from './ProductCard';
import { coffeeProducts } from '@/data/products';

export default function ProductShowcase() {
 return (
  <section className="py-24 px-4 md:px-8 relative">
   {/* Coffee Splash Banner */}
   <motion.div
    initial={{ opacity: 0, y: -100 }}
    whileInView={{ opacity: 1, y: 0 }}
    viewport={{ once: true }}
    className="relative h-64 mb-16 rounded-3xl overflow-hidden"
   >
    <div className="absolute inset-0 bg-gradient-to-r from-[#3D2418] via-[#4D3428]
to-[#3D2418]" />
    <img
     src="/coffee/splash-banner.jpg"
     alt="Coffee Splash"
     className="absolute inset-0 w-full h-full object-cover mix-blend-overlay opacity-60"
    />

    {/* Floating Coffee Beans */}
    {[...Array(8)].map((_, i) => (
     <motion.div
      key={i}
      initial={{ y: 0 }}
      animate={{
       y: [0, -20, 0],
       x: [0, Math.random() * 40 - 20, 0],
       rotate: [0, 360]
      }}
```

```
            transition={{
              repeat: Infinity,
              duration: 3 + Math.random() * 2,
              delay: i * 0.3
            }}
            className="absolute w-8 h-8 opacity-40"
            style={{
              left: `${10 + i * 12}%`,
              top: `${20 + Math.random() * 40}%`
            }}
          >
            <img src="/coffee/bean.png" alt="Coffee Bean" className="w-full h-full
object-contain" />
          </motion.div>
        ))}
      </motion.div>

      {/* Product Grid */}
      <div className="max-w-7xl mx-auto">
        <motion.h2
          initial={{ opacity: 0, y: 30 }}
          whileInView={{ opacity: 1, y: 0 }}
          viewport={{ once: true }}
          className="text-6xl md:text-7xl font-['Playfair_Display'] font-bold text-center
text-[#F5E6D3] mb-16"
        >
          Our Signature Blends
        </motion.h2>

        <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-8">
          {coffeeProducts.map((product, index) => (
            <ProductCard key={product.id} product={product} index={index} />
          ))}
        </div>
      </div>
    </section>
  );
}
```

---

components/FinalCTA.tsx

```
'use client';

import { motion } from 'framer-motion';

export default function FinalCTA() {
```

```
  return (
    <section className="py-32 px-4 relative overflow-hidden">
      {/* Animated Background */}
      <div className="absolute inset-0 bg-gradient-to-b from-[#1A0F0A] to-[#2D1810]" />
      <motion.div
        animate={{
          scale: [1, 1.2, 1],
          opacity: [0.3, 0.5, 0.3]
        }}
        transition={{ repeat: Infinity, duration: 8 }}
        className="absolute top-1/2 left-1/2 -translate-x-1/2 -translate-y-1/2 w-96 h-96
bg-[#4F9C8F]/20 rounded-full blur-3xl"
      />

      <div className="max-w-4xl mx-auto text-center relative z-10">
        <motion.h2
          initial={{ opacity: 0, y: 30 }}
          whileInView={{ opacity: 1, y: 0 }}
          viewport={{ once: true }}
          className="text-5xl md:text-7xl font-['Playfair_Display'] font-bold text-[#F5E6D3]
mb-6"
        >
          Find the Perfect Coffee for You
        </motion.h2>

        <motion.p
          initial={{ opacity: 0, y: 20 }}
          whileInView={{ opacity: 1, y: 0 }}
          viewport={{ once: true }}
          transition={{ delay: 0.2 }}
          className="text-lg md:text-xl text-[#C9B8A0] mb-12 font-['Inter']"
        >
          Experience the art of coffee craftsmanship
        </motion.p>

        <motion.button
          initial={{ opacity: 0, scale: 0.8 }}
          whileInView={{ opacity: 1, scale: 1 }}
          viewport={{ once: true }}
          transition={{ delay: 0.4 }}
          whileHover={{ scale: 1.05 }}
          whileTap={{ scale: 0.95 }}
          className="px-16 py-5 bg-gradient-to-r from-[#4F9C8F] to-[#3D8B7F] text-white
rounded-full text-xl font-semibold font-['Inter'] shadow-2xl hover:shadow-[#4F9C8F]/40
transition-shadow"
        >
          Explore Full Menu
        </motion.button>
```

```
      {/* Decorative Sparkle */}
      <motion.div
        animate={{ rotate: 360, scale: [1, 1.2, 1] }}
        transition={{ repeat: Infinity, duration: 4 }}
        className="mt-12 text-[#D4A574] text-4xl"
      >
        ✦
      </motion.div>
    </div>
  </section>
);
}
```

E. Main Page

app/page.tsx

```
'use client';

import { useEffect } from 'react';
import HeroCanvasAnimation from '@/components/HeroCanvasAnimation';
import ProductShowcase from '@/components/ProductShowcase';
import FeatureSection from '@/components/FeatureSection';
import FinalCTA from '@/components/FinalCTA';

export default function Home() {
  useEffect(() => {
    // Ensure smooth scroll behavior
    document.documentElement.style.scrollBehavior = 'smooth';
  }, []);

  return (
    <main className="bg-[#1A0F0A] min-h-screen">
      {/* Hero: Scroll-Triggered Canvas Animation */}
      <HeroCanvasAnimation />

      {/* Product Showcase Section */}
      <ProductShowcase />

      {/* Feature Highlights Section */}
      <FeatureSection />

      {/* Final Call-to-Action */}
      <FinalCTA />
    </main>
```

```
  );
}
```

✅ KEY REQUIREMENTS CHECKLIST

Hero Canvas Animation:

- ✅ Loads frames from `/public/frames/frame_0.webp` to `/public/frames/frame_N.webp`
- ✅ **Bi-directional scroll** (forward on scroll down, reverse on scroll up)
- ✅ **Anti-gravity physics** using scroll velocity
- ✅ Smooth `useSpring` interpolation (stiffness: 100, damping: 30)
- ✅ Preload ALL frames with progress bar
- ✅ Responsive canvas scaling (contain fit logic)
- ✅ Text overlays with scroll-based fade in/out
- ✅ Scroll indicator that fades out
- ✅ Proper cleanup (event listeners, canvas context)

Product Sections:

- ✅ Match reference image design aesthetic
- ✅ Coffee brown color palette (`#2D1810`, `#3D2820`, etc.)
- ✅ Product cards with ratings, prices, add-to-cart buttons
- ✅ Feature grid with center coffee cup image
- ✅ Hover animations (scale, glow effects)
- ✅ Fully responsive (mobile → desktop)
- ✅ Teal accent for CTAs (`#4F9C8F`)
- ✅ Scroll reveal animations with Framer Motion

Performance & Polish:

- ✅ TypeScript throughout (100% type-safe)
- ✅ Static export compatible (`output: 'export'`)
- ✅ 60fps smooth animations
- ✅ Custom scrollbar styling
- ✅ Loading state before animation
- ✅ No flicker/stutter during scroll
- ✅ Proper font loading (Google Fonts)

📦 HOW TO USE

1. Setup Frames:

```
# Extract your frame zip file
# Rename frames: frame_0.webp, frame_1.webp, ..., frame_119.webp
# Place in: /public/frames/

# Adjust TOTAL_FRAMES constant in HeroCanvasAnimation.tsx
```

2. Install & Run:

```
npm install
npm run dev
```

3. Build for Production:

```
npm run build
# Deploy the /out folder to Netlify/Vercel
```

🎨 CUSTOMIZATION NOTES

- **Frame Count**: Update `TOTAL_FRAMES` in `HeroCanvasAnimation.tsx`
- **Scroll Duration**: Adjust `h-[500vh]` in hero container (higher = slower scroll)
- **Anti-Gravity Strength**: Modify `yOffset` transform values `[15, 0, -15]`
- **Text Content**: Edit text overlays directly in `HeroCanvasAnimation.tsx`
- **Products**: Update `data/products.ts` with your coffee offerings
- **Colors**: Modify Tailwind config or use CSS variables